Retail Analytics & Machine Learning Project

Churn Prediction, Future Monetary Value Estimation, and Customer Segmentation

Author: *Chayma Dallel*

(GI2 S3)

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Abstract

This project aims to predict and analyze customer behavior for a retail website through three complementary lenses: (i) a classification task to predict churn (at-risk customers), (ii) a regression task to estimate future monetary value (using `MonetaryTotal` as a proxy), and (iii) a clustering task to segment customers into homogeneous groups.

The proposed solution is structured as an industrial-grade pipeline: data ingestion, preprocessing, training, evaluation, and deployment via a Web application (Flask) and/or a Streamlit interface.

The dataset consists of 4,372 customers and 52 variables (numerical and categorical). The modules generate EDA reports, aligned train/test sets, serialized models, and comprehensive visualizations.

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1 Context and Objectives

1.1 Business Problem

In a retail context, the ability to anticipate customer loss (churn) and measure future value is critical. A data-driven approach enables:

- Retention prioritization via targeted campaigns for high-risk customers.
- Value optimization through upsell and cross-sell opportunities for high-value customers.
- Customer segmentation for personalized marketing and tailored offers.

1.2 Machine Learning Objectives

The project encompasses three primary tasks:

1. Classification (Churn): predict the binary target variable `Churn`.
2. Regression (Monetary): predict the continuous variable `MonetaryTotal`.
3. Clustering (Segmentation): segment customers using the KMeans algorithm.

2 Data Overview

2.1 Source and Dimensions

- Dataset: `retail_customers_COMPLETE_CATEGORICAL.csv`
- Row count: 4,372
- Column count: 52

2.2 Data Quality and EDA Findings

Exploratory Data Analysis (EDA) and generated reports highlight the following:

- Duplicates: 0
- Missing values: predominantly found in `Age` and `AvgDaysBetweenPurchases`.
- Suspicious values: observed entries such as `-1`, `99`, `999`, `Unknown`, `NA`, and empty strings in specific variables.

2.3 Key Variables

Typical variables oriented around RFM (Recency, Frequency, Monetary) and customer profiles:

- RFM: `Recency`, `Frequency`, `MonetaryTotal`.
- Demographics: `Age`, `Gender`, `Region`.
- Engagement: `SatisfactionScore`, preferences, and loyalty status.

3 Software Architecture

3.1 Modular Organization

The project is structured into modules aligned with the ML lifecycle:

- Preprocessing: `src/preprocessing.py`
- Training: `src/train_model.py`
- Evaluation: `src/evaluation.py`
- Prediction: `src/predict.py`
- Utilities: `src/utils.py`
- Feature engineering: `src/features.py` (Scikit-learn transformer)
- Web deployment: `app.py` + HTML templates

3.2 Directory Structure

- `data/raw/` — Raw datasets.
- `data/train_test/` — Transformed sets (`X_train_pca.csv`, etc.).
- `data/predictions/` — Output files (`predictions.csv`).
- `models/` — Serialized models (`.pkl`) and input schema (`raw_schema.json`).
- `reports/` — Logs, EDA reports, plots, VIF analysis, etc.

4 Preprocessing Pipeline

4.1 Core Stages

The preparation pipeline is designed for production reusability: a single transformation chain is serialized and applied identically during both training and inference.

1. Suspicious value cleaning: replacement of anomalies with `NaN`.
2. Feature engineering:
   - `RegYear` and `RegMonth` extracted from `RegistrationDate`.
   - $\texttt{MonetaryPerDay} \approx \frac{\texttt{MonetaryTotal}}{\texttt{Recency}+1}$.
   - $\texttt{AvgBasketValue} \approx \frac{\texttt{MonetaryTotal}}{\texttt{Frequency}}$.
   - `TenureRatio` (if `CustomerTenure` is available).
3. Age handling: values outside the range 18–81 are set to `NaN`.
4. Imputation strategy: median for numerical variables; mode for categorical variables.
5. Encoding: OneHotEncoder with `handle_unknown=ignore`.
6. Scaling: StandardScaler for numerical features.
7. Dimensionality reduction: PCA with `n_components=10`.

4.2 Data Leakage Management

Certain variables may encapsulate information too closely related to the target (e.g., `ChurnRisk`). These are removed from the pipeline via a specific `drop_cols` list to ensure model integrity.

4.3 Production Robustness: Input Schema

A `models/raw_schema.json` file saves the list of expected columns for inference. In the case of partial input (Web forms), the DataFrame is reindexed: missing columns are filled with `NaN` and processed through the imputer.

5 Models and Methodology

5.1 Train/Test Split

The data is split 80% training / 20% testing with stratification on the churn target:

$$(X_{train},\ X_{test},\ y^{cls}_{train},\ y^{cls}_{test},\ y^{reg}_{train},\ y^{reg}_{test})$$

5.2 Classification: Churn Prediction

Model selection. XGBoost Classifier is utilized for its superior performance on tabular data and its ability to capture non-linear relationships.

Output. `Churn_Probability` gives a probability in $[0, 1]$; `Churn_Predicted` applies a default threshold of 0.5 (adjustable).

Metrics. Accuracy, AUC ROC, Precision, Recall, and F1-score.

5.3 Regression: Monetary Value Estimation

Model selection. RandomForestRegressor is employed to model non-linear relationships while maintaining resistance to noise.

Metrics. RMSE, R^2, and visual analysis via actual vs. predicted scatter plots.

5.4 Clustering: Customer Segmentation

Model selection. KMeans is applied to segment customers into homogeneous groups within the PCA-transformed space.

Determining k. The Elbow Method is computed for $k \in [2, 10]$. The optimal value is selected via heuristic or the elbow point.

Output. `cluster` — segment identifier (0, 1, 2, ...).

6 Evaluation and Optimization

6.1 Hyperparameter Tuning

The `evaluation.py` module performs hyperparameter optimization for XGBoost via GridSearchCV.

Optimization metric. The primary recommended score is ROC AUC, as churn is often imbalanced and the cost of errors is asymmetric.

6.2 Consistency Checks

The scripts include validation steps to prevent:

- Mismatch between `X` and `y` sizes.
- Double-transformation of data.
- Schema inconsistencies between training and inference phases.

7 Prediction (Inference)

7.1 Test Mode

In test mode, models are applied to `X_test_pca.csv`. A `predictions.csv` file is saved, containing:

- Churn probability.
- Predicted churn vs. actual churn.
- Predicted monetary value vs. actual value.
- Assigned cluster.

7.2 Sample Mode (New Customers)

Sample mode simulates new customers by drawing from raw data and applying:

$$X_{raw} \xrightarrow{\text{pipeline}} X_{pca} \xrightarrow{\text{models}} \hat{y}$$

This ensures the transformation chain remains identical to the one used during training.

8 Deployment: Web Application

8.1 Flask Framework

A Flask application exposes three primary routes:

- `/predict_classification`
- `/predict_regression`
- `/predict_clustering`

8.2 User Interface

The HTML form allows users to input key variables such as `Age` and `SatisfactionScore`. The pipeline fills in missing variables via the schema, then applies imputation and transformations.

8.3 Limitations

For maximum production accuracy, it is recommended to expose more input variables in the form and to connect the API to a database containing the full history of customer features.

9 Results and Deliverables

9.1 Generated Files

- Data: `X_train_pca.csv`, `X_test_pca.csv`, `y_train.csv`, etc.
- Models: `churn_classifier.pkl`, `monetary_regressor.pkl`, `customer_clusters.pkl`.
- Pipeline: `preprocessing_pipeline.pkl`.
- Schema: `raw_schema.json`.
- Reports: `data_report.txt`, EDA plots, elbow curve, regression scatter plots.

9.2 Visualizations

The following charts are typically produced:

- Histograms (RFM, Age, etc.).
- Correlation heatmaps.
- Regression scatter plots (actual vs. predicted).
- Elbow curve (KMeans).
- 2D cluster projections (PC1 vs. PC2).

10 Discussion: Technical Choices, Limits, and Future Work

10.1 Technical Choices

- Scikit-learn Pipeline: ensures reproducibility and consistency.
- XGBoost: high performance on tabular datasets.
- RandomForestRegressor: a robust and interpretable baseline.
- KMeans: simple and actionable segmentation.

10.2 Limitations

- PCA reduces interpretability as features are transformed into principal components.
- The 0.5 threshold is not optimized for business costs and should be calibrated based on financial impact.
- KMeans assumes spherical clusters and relies on Euclidean distance.

10.3 Recommended Improvements

1. Imbalance management: class weights, SMOTE, or Focal Loss.
2. Probability calibration: Platt scaling or isotonic regression.

3. Explainability: implementation of SHAP to explain churn drivers.
4. Optimal k selection: silhouette score or Davies-Bouldin index.
5. Monitoring: data drift and performance drift logs.
6. CI/CD: unit tests for schema consistency and pipeline reproducibility.

11 Execution Guide

11.1 Recommended Workflow

```python
# 1) Preprocessing + Artifact saving
python src/preprocessing.py

# 2) Model Training
python src/train_model.py

# 3) Prediction (Validation on test set)
python src/predict.py --mode test

# 4) Launch Flask App
python app.py
```

Listing 1: Step-by-step execution commands

11.2 Reproducibility

- Set `random_state=42` in all splits and models.
- Maintain library versions via `requirements.txt`.
- Always serialize the pipeline along with the models.

12 Appendix: Code Snippets

12.1 Model Serialization

```python
from utils import save_model

# Exporting the trained classifier
save_model(clf, "models/churn_classifier.pkl")
```

Listing 2: Exporting a trained classifier

12.2 Pipeline Concept

```python
pipe = Pipeline(steps=[
    ("fe", FeatureEngineer()),
    ("prep", ColumnTransformer([...])),
    ("pca", PCA(n_components=10))
])
```

Listing 3: Scikit-learn pipeline structure

Conclusion

This project provides a comprehensive, modular, and deployable solution for churn prediction, monetary value estimation, and customer segmentation in the retail sector. Through standardized pipelines, model serialization, and an accessible Web interface, the system ensures a smooth transition from prototype to production. Future enhancements focusing on explainability (SHAP) and business-cost optimization will further increase the strategic value of this tool.